THE AUSTRALIAN NATIONAL UNIVERSITY

Mid-Semester Examination – August 31st, 2021

COMP3320/6464/HONS
HIGH PERFORMANCE SCIENTIFIC COMPUTATION

- Study period: 15 minutes
- Time allowed: 1 hour and 45 minutes (including submission time to Wattle)
- Permitted materials: open-book exam
- Exam questions total 100 marks
- All students answer all questions
- Clarity and conciseness in answers will be highly valued
- Marks may be lost for supplying irrelevant information

Topic 1: Floating point arithmetic [25 total marks]

(a) Your machine uses a binary floating-point representation ($\beta = 2$) with precision $p = 2$, and whose smallest and largest exponents are $L = -3$ and $U = 3$, respectively. Answer the following questions and justify your answer:

(i) How many normalized floating-point numbers can be represented by your machine? **[3 marks]**

(ii) What are the three smallest (non-zero) normalized positive numbers? **[3 marks]**

(iii) What are subnormal numbers, and in which arithmetic range would they go? **[2 marks]**

(iv) How many subnormal numbers can be represented by your machine? **[1 mark]**

(b) The following code illustrates the effects of both approximation and rounding errors:

```
#include <stdio.h>
#include <math.h>
int main(void)
{ float h,x;
  int i;
  x=9.87654321;
  for (i=0, h=0.1; i<13; i++, h*=0.1){
       printf("%16.14f   %8.1e   %16.14f \n", 2.0*(1.0+x), h,
              ((2.0*(x+h)+(x+h)*(x+h))-(2*x+x*x))/h);
  }
  return 0;
}
```

It computes the derivative $2x+x^2$ both analytically and using the following finite difference method:

$$\frac{d(2x+x^2)}{dx} = 2(1+x) \approx \frac{2(x+h)+(x+h)^2-(2x+x^2)}{h}$$

where h is the step size used in the finite difference. The code uses `x=9.87654321` and various step sizes. It gives the following output:

```
21.75308609008789    1.0e-01   21.85310331205870
21.75308609008789    1.0e-02   21.76284838682819
21.75308609008789    1.0e-03   21.75712735394707
21.75308609008789    1.0e-04   21.72470305722672
21.75308609008789    1.0e-05   19.83642808647295
21.75308609008789    1.0e-06   13.35144198127987
21.75308609008789    1.0e-07   -38.14697763147228
21.75308609008789    1.0e-08   -381.46976276219249
21.75308609008789    1.0e-09   -3814.69779702855431
21.75308609008789    1.0e-10   -38146.97744088980835
21.75308609008789    1.0e-11   -381469.76117400545627
```

```
21.75308609008789     1.0e-12    -3814697.69445813260972
21.75308609008789     1.0e-13    -38146975.91060535609722
```

(i) Explain how this program illustrates both rounding error and approximation error.
[5 marks]

(ii) For any given value of h and without changing the value of h, state how you would modify the code to: (A) Decrease the effect of rounding errors; (B) decrease the effect of approximation errors.
[5 marks]

(iii) For $\mathrm{h} = 10^{-7}$ write the expression for the relative error of the numerical derivative and evaluate it.
[4 marks]

(iv) Explain why for $\mathrm{h} \leq 10^{-7}$ the numerical derivative values are negative.
[2 marks]

Topic 2: CPU architecture [25 total marks]

(a) Over time there has been a tendency for all microprocessors to increase the number of stages in their pipeline: i) Give one advantage of long pipelines; ii) Give at least one (fundamentally different) disadvantage of long pipelines; iii) State how microprocessor designers have sought to minimize the disadvantages of long pipelines. **[5 marks]**

(b) Data dependencies impact on the ability to pipeline. Give a few lines of code containing a for loop where there is a data dependency between iterations of the loop. Clearly indicate the dependency. **[5 marks]**

(c) You are running on a machine with:

- Load/store latency of 3 cycles to L1 cache.
- Floating point add or multiply latency of 2 cycles.
- Superscalar architecture capable of one load, one store, one floating point addition, one floating point multiplication and 3 integer operations per clock cycle.

Consider the following loop in which all variables are independent, and `x` and `y` are floating point variables, and `a` and `b` are floating-point constants.

```
for (i=0; i<2*N; i++)
    y[i] += a*x[i]+b;
```

Assume that there is no aliasing and no loop overhead (i.e., branching and increment of counter `i` are for free).

(i) What is the minimum number of clock cycles that it will take to execute one loop iteration, when the compiler has not performed loop unrolling and pipelining? Give an instruction schedule, showing every clock cycle as a separate line, to explain your answer.

[5 marks]

(ii) What is the speedup achieved when the loop is unrolled 2 times? Include the new code (in the format included above) and a new instruction schedule in your answer.
[5 marks]

(iii) What is the speedup achieved with respect to the number of cycles required by (i) when load instructions are accelerated by a pipeline of depth 3 and the loop is unrolled 4 times?
[5 marks]

Topic 3: Memory Architecture [25 marks]

a) Your machine has a CPU with a 1 KB four-way associative L1 cache. The L1 cache line size is 64 bytes. You run the following vector triad

```
for(i=0; i<32; i++)
  for(j=0; j<N; j+=32)
    A[i+j] = B[i+j]+C[i+j]*D[i+j];
```

where `N` is large and `A`, `B`, `C` and `D` are double precision arrays.

(i) Assuming no pipelining or loop unrolling is performed, how many loads, stores and floating-point instructions are executed per iteration? **[5 marks]**

(ii) What is the average read miss rate per iteration? Carefully justify your answer. Note: the read miss rate is defined as the ratio between the number of read misses and the sum of the number of read misses and read hits. **[5 marks]**

(b) Processors use caches to improve their performance.

(i) Explain what spatial and temporal locality in the context of caches is. **[2 marks]**

(ii) Provide one example of C code that does not take advantage of temporal locality. Provide a detailed motivation for your choice. **[3 marks]**

(c) Shown below are three functionally identical C routines to perform a matrix-matrix multiplication.

```
//Routine #1
for(j = 0; j < N; j++)
  for(k = 0; k < N; k++){
     Bkj = B[j+N*k];
     for(i = 0; i < N; i++)
       C[j+N*i] += A[k+N*i]*Bkj;
  }

//Routine #2
for(k = 0; k < N; k++)
  for(i = 0; i < N; i++){
```

```
      Aik = A[k+N*i];
      for(j = 0; j < N; j++)
        C[j+N*i] += Aik*B[j+N*k];
  }

//Routine #3
for(i = 0; i < N; i++)
  for(k = 0; k < N; k++){
      Aik = A[k+N*i];
      for(j = 0; j < N; j++)
        C[j+N*i] += Aik*B[j+N*k];
  }
```

where A, B and C are double precision arrays.

(i) Rank the routines in order of decreasing performance and justify your ranking.
[6 marks]

(ii) How could you improve the performance of the top-performing routine? Provide a short code snippet and justify your answer.
[4 marks]

Topic 4: Performance Measurement and Optimization [25 marks]

(a) Rewrite the following C code snippets to improve serial performance on an Intel Xeon processor (the one you use on Gadi) without changing code functionality, and reason why your change(s) improves performance.

(i) Vector operations

```
M = N/2;
for(i = 0; i < N; i++)
  a[i] = b[i]*c[i];
for(j = 0; j < M; j++)
  z[j] = b[j]+y[j];
```

[5 marks]

(ii) Matrix transpose

```
for(i = 0; i < N; i++)
  for(j = 0; j < N; j++)
    a[j][i] = b[i][j];
```

[5 marks]

(b) In timing your application code, when would you use process time and when would you use wall clock time? **[5 marks]**

(c) The process time is usually broken down into “user” and “system” components. What is included in each of these components? **[2 marks]**

(d) You are given the following C code.

```c
void mycode(double a[], double b[], double c[], int N){
int i, j;
int count;

count=0;
for (i=0; i<N; i++){
  a[i] = 0.0;
  count++;
}
printf("end loop 1 %d\n",count);

count=0;
for (i=0; i<N; i++){
  for (j=i; j<N; j++){
    a[i] = a[i] + b[j];
    count++;
  }
}
printf("end loop 2 %d\n",count);

count=0;
for (i=0; i<N; i++){
  for (j=1; j<N*N; j*=2){
    a[i] = a[i] + c[i+j]*2.0;
    count++;
  }
}
printf("end loop 3 %d\n",count);
return;
}
```

(i) Write general expressions for the scaling of each loop nest as a function of `N`. **[6 marks]**

(ii) If N has value 5, what `count` values are printed for each loop? **[3 marks]**